

Fast Food Marketing Campaign

Goal of the Test

To determine which of the three marketing promotions generates the highest sales performance across store locations.

The [dataset wa_marketing_campaign](#) contains weekly sales data for different store locations that were assigned one of three promotional campaigns. We aim to identify which campaign performs best and should be implemented more broadly by comparing the sales results of each promotion.

Target Metric

Average sales in thousands per location. Aggregated across the four test weeks. It measures the financial performance of each promotion. The goal of the marketing campaign is to increase sales. Comparing the average sales across promotions will determine which campaign performs best.

Assumptions of the Two-Sample t-Test

Assumption	Description
Random	Stores were assigned to promotions as part of a marketing experiment. We assume the assignment was random.
Independent	Each store operates independently, and sales from one store do not affect another store. Sample size is less than 10% of total possible stores.
Normal	Sample sizes are 43, 47, and 47 (all greater than 30). By the Central Limit Theorem, the sampling distribution of the mean is approximately normal.

Since all three assumptions are satisfied, the use of a two-sample t-test is appropriate.

Calculations

Statistical Test: Promotion 1 and Promotion 2

A two-sample t-test is conducted at a 99% confidence level to compare Promotion 1 and Promotion 2.

1. Promotion 1 mean sales: 232.40

2. Promotion 2 mean sales: 189.32

3. Difference in means: 43.08

4. p-value: 0.00128

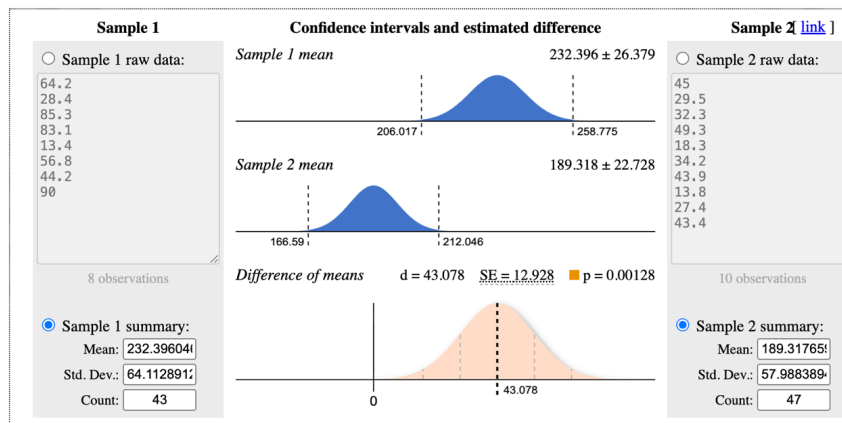
Statistic	t-score
Probability	$P(T \leq t)$
Degrees of freedom	88
t-score	2.63286
$P(T \leq 2.63286)$	0.995

The p-value is lower than 0.01. We reject the null hypothesis. The results show that Promotion 1 generates significantly higher sales than Promotion 2.

Evan's Awesome A/B Tools ([home](#)):

[Sample Size Calculator](#) | [Chi-Squared Test](#) | [Sequential Sampling](#) | [2 Sample T-Test](#) | [Survival Times](#) | [Count Data](#)

Question: Does the average value differ across two groups?



Verdict: Sample 1 mean is greater

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

Statistical Test: Promotion 1 and Promotion 3

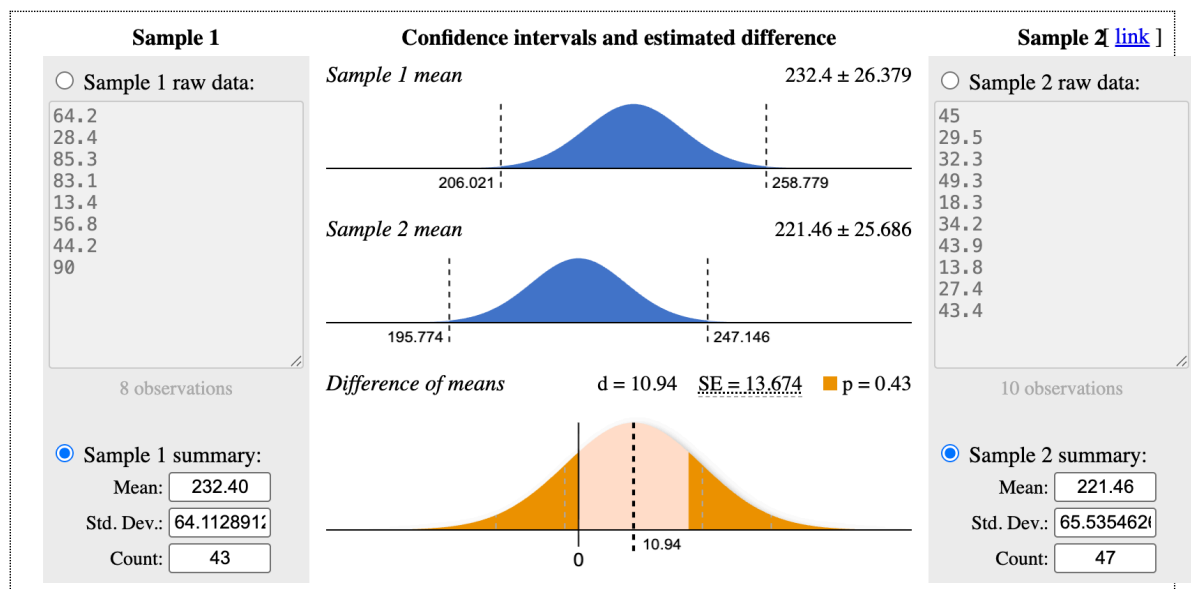
A two-sample t-test is conducted at a 99% confidence level to compare Promotion 1 and Promotion 3.

1. Promotion 1 mean sales: 232.40
2. Promotion 3 mean sales: 221.46
3. Difference in means: 10.94
4. p-value: 0.43

The p-value is greater than 0.01. We fail to reject the null hypothesis. There is no statistically significant difference between Promotion 1 and Promotion 3.

[Sample Size Calculator](#) | [Chi-Squared Test](#) | [Sequential Sampling](#) | [2 Sample T-Test](#) | [Survival Times](#) | [Count Data](#)

Question: Does the average value differ across two groups?



Statistical Test: Promotion 2 and Promotion 3

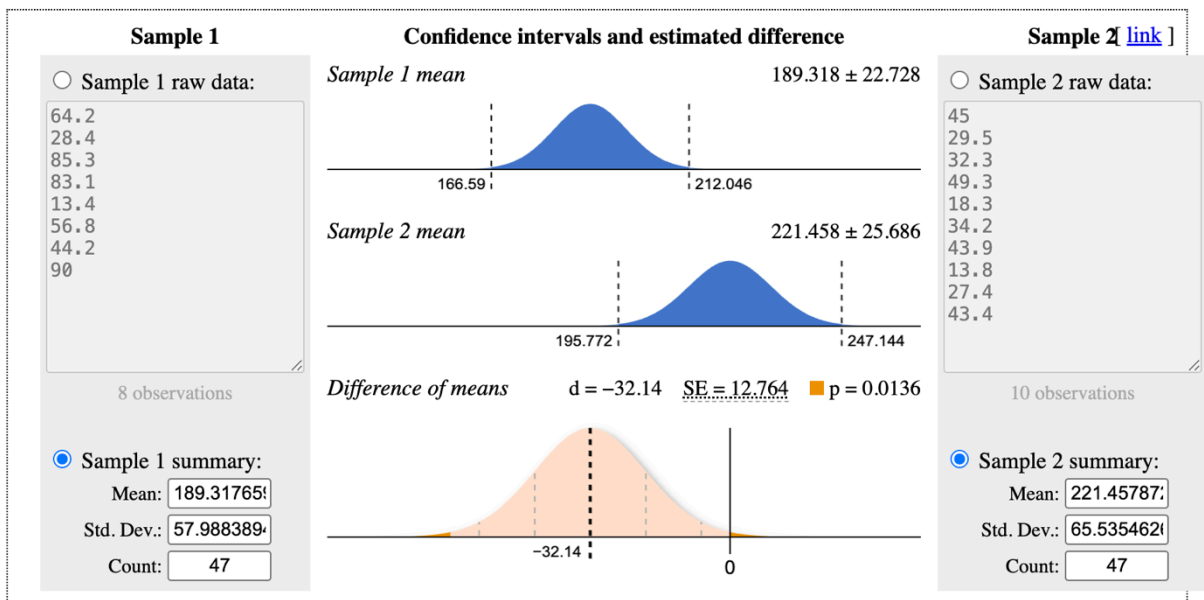
A two-sample t-test is conducted at a 99% confidence level to compare Promotion 2 and Promotion 3.

1. Promotion 2 mean sales: 189.32
2. Promotion 3 mean sales: 221.46
3. Difference in means: -32.14
4. p-value: 0.0136

The p-value is greater than 0.01. We fail to reject the null hypothesis. There is no statistically significant difference between Promotion 2 and Promotion 3.

[Sample Size Calculator](#) | [Chi-Squared Test](#) | [Sequential Sampling](#) | [2 Sample T-Test](#) | [Survival Times](#) | [Count Data](#)

Question: Does the average value differ across two groups?



Verdict: No significant difference

Hypothesis: ☒ $d = 0$ ☐ $d \leq 0$ ☐ $d \geq 0$

Confidence: 99%

Decision

Based on the statistical analysis conducted at the 99% confidence level, Promotion 1 significantly outperforms Promotion 2. The difference in average sales between these two promotions is statistically significant ($p = 0.00128$).

But, there is no statistically significant difference between Promotion 1 and Promotion 3, nor between Promotion 2 and Promotion 3 at the 99% confidence level.

Promotion 1 has the highest average sales overall. But the difference between Promotion 1 and Promotion 3 is not statistically significant. Therefore, both Promotion 1 and Promotion 3 appear to perform similarly. And Promotion 2 performs worse than Promotion 1.

Based on this analysis, Promotion 1 is recommended as the preferred marketing campaign because it shows the highest average sales and significantly outperforms Promotion 2.

Three pairwise comparisons were conducted. There is an increased risk of Type I error (false positives). To account for the multiple comparisons problem, a stricter confidence level of 99% ($\alpha = 0.01$) was used instead of the conventional 95% level.

Recommendation

Promotion	Performance Level	Risk Level	Business Recommendation
1	Highest average sales	Low risk (significant win vs P2)	Roll out broadly across markets
2	Lowest average sales	High risk	Discontinue or redesign
3	Comparable to P1	Medium uncertainty	Consider cost comparison before rollout

Limitations

The dataset only includes total sales by store and promotion, without additional contextual information. The results could have been more robust if we had data, like: store size, location demographics, marketing costs per promotion, customer traffic, or historical baseline sales before the campaign. These variables would allow for analysis, such as controlling for external factors or calculating return on investment (ROI).

Next Steps

1. Segmenting results by market size, store age, or geographic region to determine whether certain promotions perform better in specific contexts.
2. A longer testing period could also improve statistical power and reduce uncertainty between Promotions 1 and 3.
3. Analyzing cost data would allow calculation of return on investment (ROI) to determine the most profitable promotion.

Appendix

1. [Two-Sample T-Test \(Evan's Awesome A/B Tools\)](#)
2. [T Distribution Calculator](#)
3. SQL Queries
 - a. Step 1: Total 4-week sales for one store under one promotion

<https://console.cloud.google.com/bigquery?sq=147855269776:d6f9fdcc5a2047329a56191b5dde78f9>

- b. Step 2: Calculate Summary Statistics

<https://console.cloud.google.com/bigquery?sq=147855269776:67cd3cc8d9c344ba97c7b903f12df055>

Determining the Critical Value for Promotion 1 and 2:

Degree of Freedom, $df=43+47-2=88$

For a 99% two-tailed test:

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

For two-tailed, the cumulative probability is: $1 - 0.005 = 0.995$

Critical value for a 99% two-tailed test:

$$t^{\text{crit}} = 2.63286$$

Finding CI:

$$99\% \text{ Confidence Interval} = 43.078 \pm (2.63286 \times 12.928) = [9.05, 77.11]$$

The interval does not include 0. We can conclude that the difference is statistically significant at the 99% confidence level.

Determining the Critical Value of Promotion 1 and 3:

$$\text{Degree of Freedom, } df = 43 + 47 - 2 = 88$$

For a 99% two-tailed test:

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

For two-tailed, the cumulative probability is: $1 - 0.005 = 0.995$

Critical value for a 99% two-tailed test:

$$t^{\text{crit}} = 2.63286$$

Finding CI:

$$99\% \text{ Confidence Interval} = 10.938 \pm (2.63286 \times 13.674) = [-25.05, 46.93]$$

The interval includes 0. We cannot conclude that the difference is statistically significant at the 99% confidence level.

Determining the Critical Value of Promotion 2 and 3:

Degree of Freedom, $df = 47 + 47 - 2 = 92$

For a 99% two-tailed test:

$$\alpha = 0.01$$

$$\alpha/2 = 0.005$$

For two-tailed, the cumulative probability is: $1 - 0.005 = 0.995$

Using $df = 92$ in the t-distribution calculator:

Critical value for a 99% two-tailed test:

$$t_{crit} = 2.63$$

Finding CI:

$$99\% \text{ Confidence Interval} = -32.14 \pm (2.63 \times 12.764) = [-65.70, 1.42]$$

The interval includes 0. We cannot conclude that the difference is statistically significant at the 99% confidence level.